

Here are your notes based on the two videos about Standard Deviation and Standard Error.

Lecture Notes: Standard Deviation vs. Standard Error

These two concepts are frequently confused but measure two very different things: **Standard Deviation (SD)** measures the *spread of your data*, while **Standard Error (SE)** measures the *precision of a statistic* (like the mean).

1. The Core Difference (SD vs. SE)

Standard Deviation (SD)

- **What it is:** A measure of the **variation *within* a single set of measurements**.
- **Analogy:** You weigh 5 mice. The SD quantifies how much the individual weights of those 5 mice are spread out around their average (mean) weight.
- **What it tells you:** "How much do my individual data points vary?"
- **When to plot it:** Almost always. Graphs are usually intended to describe the data you actually collected. An SD error bar shows the *spread of the data*.

Standard Error (SE)

- **What it is:** The **standard deviation of a statistic**. Most often, it's the **Standard Error of the Mean (SEM)**.
- **Analogy:** You do the 5-mice experiment five *separate times* (using new mice each time). You now have 5 different mean weights. The **Standard Error** is the **standard deviation of those 5 means**.
- **What it tells you:** "How precise is my sample mean as an estimate of the *true population mean*?" or "If I repeated this experiment, how much would I expect the *mean* to jump around?"
- **When to plot it:** When you want to show the *precision of your mean* or compare means (e.g., with t-tests). An SE error bar (which is smaller than SD) shows the *confidence in your mean*.

The Source of Confusion: You can *estimate* the Standard Error from a **single experiment**, even though it *describes* the variation from *multiple* hypothetical experiments.

2. How to Calculate Standard Error (From a Single Sample)

You don't need to actually repeat your experiment 1000 times. You can *estimate* the SE from your one sample.

Method 1: The Formula (For the Mean Only)

For the **Standard Error of the Mean (SEM)**, there is a simple formula:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

- s : Your sample's **Standard Deviation**.
- n : Your **sample size** (e.g., 5 mice).

Method 2: Bootstrapping (For Everything Else)

What if you want the Standard Error of the **median**, or the 75th percentile, or a statistic that has no simple formula? You use **bootstrapping**.

Bootstrapping is a computational method to *simulate* repeating the experiment.

The 5-Step Process:

1. **Start with your sample:** You have one sample of n measurements (e.g., your 5 mice weights).
2. **Create a "Bootstrap Sample":** Create a *new* sample of size n by randomly picking values from your original sample *with replacement*.
 - "With replacement" means you can pick the same value multiple times.
 - Original Sample:* [10g, 12g, 11g, 13g, 10g]
 - A Bootstrap Sample might be:* [12g, 10g, 10g, 13g, 12g]
3. **Calculate the Statistic:** Calculate your statistic (e.g., the median) for this new bootstrap sample.
4. **Repeat:** Go back to Step 2 and repeat this process thousands of times (e.g., 10,000 times), creating thousands of new bootstrap statistics.
5. **Calculate the SE:** You now have a large list of 10,000 bootstrap medians. The **standard deviation of this new list** is your **Standard Error of the Median**.

Summary

Feature	Standard Deviation (SD)	Standard Error (SE)
What it Measures	Spread of the data	Precision of a statistic (like the mean)
Question Answered	"How spread out are my individual measurements?"	"How precise is my mean?" or "How much would the mean vary if I re-ran the study?"
Typical Use	Describing your sample	Inferring properties of the <i>population</i>
Formula (for mean)	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$	$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$
http://googleusercontent.com/youtube_content/4 http://googleusercontent.com/youtube_content/5		